Amrita School of Artificial Intelligence, Faridabad
Amrita Vishwa Vidyapeetham

Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 1.9

DMD and need of Hankelization

BTech 4th Sem

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*All Sources: Stanford University, Texas Tech University, UC Santa Barbara, etc.

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Estimating Matrix A

We do not know A. But we know

$$X' = AX$$

We want to solve for A.

Rearranging gives: $A = X'X^{-1}$

However X **is usually not square**, so its inverse does not exist.

Therefore we use the **Moore–Penrose pseudoinverse**.

Thus

$$A = X'X^{\dagger}$$

Where $X^{\dagger}$ is the **pseudoinverse of** X.

Pseudoinverse Formula

If matrix X has **full column rank**, the pseudoinverse is

$$X^{\dagger} = \left((X^T X)^{-1} X^T\right)$$

This is the **least squares solution**.

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Why Pseudoinverse?

Because the equation: $X' = AX$

may not have an exact solution.

The pseudoinverse gives the **best least-squares solution**, meaning it finds A that minimizes

$$|| X' - AX ||$$

So we obtain the **best linear approximation of the system dynamics**.

Real Life Example: Fluid Flow

Imagine measuring velocity of air around an airplane wing.

Snapshots of flow are taken using cameras.

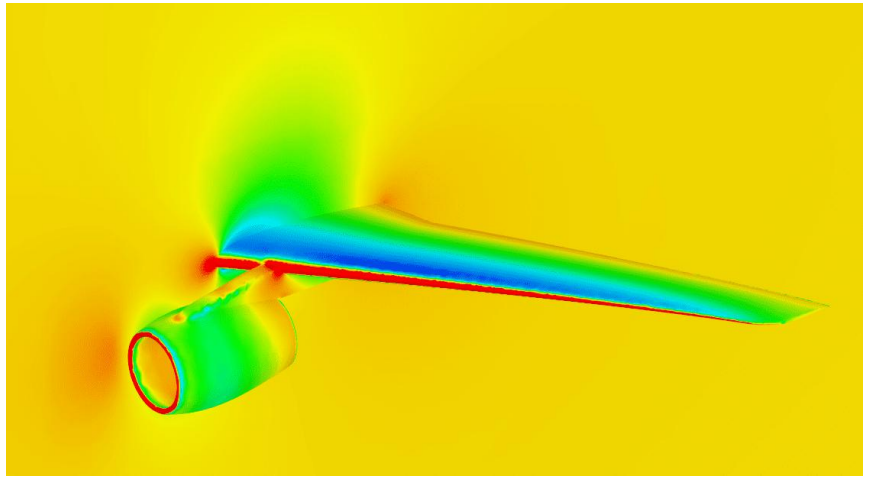
Each snapshot is a large vector x_k.

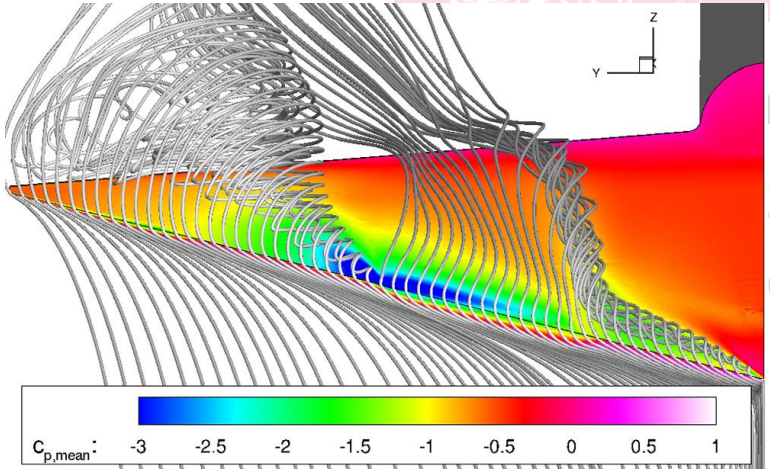
DMD analyzes these snapshots and finds:

- dominant vortex structures
- oscillation frequencies
- coherent flow patterns

This helps engineers design better wings.

Time	Image
(t_1)	airflow pattern
(t_2)	slightly different pattern
(t_3)	vortices moved
(t_4)	new flow structures





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Limitation of Standard DMD

Standard DMD assumes

$$x_{k+1} = Ax_k$$

This means system is **first-order linear**.

But many real systems are **not first-order**.

Example:

Human heartbeat depends on **multiple previous states**, not just the current one.

2. Example of Higher-Order System

Consider a system

$$x_{k+1} = 2x_k - x_{k-1}$$

This system depends on **two previous values**.

Standard DMD cannot capture this properly.

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Tricks

"We trick the system by increasing dimension."

Instead of:

$$x_k$$

We use:

$$\tilde{x}_k = \begin{bmatrix} x_k \\ x_{k-1} \\ x_{k-2} \end{bmatrix}$$

Now system becomes:

$$\tilde{x}_{k+1} = A\tilde{x}_k$$

Now DMD works!

Example of Higher-Order System

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here: $x_{k+1} = 2x_k - x_{k-1}$

we define a **new state vector**:

$$\tilde{x}_k = \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

We define:

$$\tilde{x}_{k+1} = \begin{bmatrix} x_{k+1} \\ x_k \end{bmatrix}$$

So:

$$\tilde{x}_{k+1} = \begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix}$$

We want a matrix A such that:

$$\begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix} = A \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

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We want a matrix A such that:

$$\begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix} = A \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix}$$

Assume:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then:

$$A\tilde{x}_k = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_k \\ x_{k-1} \end{bmatrix} = \begin{bmatrix} ax_k + bx_{k-1} \\ cx_k + dx_{k-1} \end{bmatrix}$$

Compare with:

$$\begin{bmatrix} 2x_k - x_{k-1} \\ x_k \end{bmatrix}$$

First row:

$$ax_k + bx_{k-1} = 2x_k - x_{k-1}$$

So:

$$a = 2, b = -1$$

Second row:

$$cx_k + dx_{k-1} = x_k$$

So:

$$c = 1, d = 0$$

FINAL MATRIX

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

FINAL RESULT

$$\tilde{x}_{k+1} = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \tilde{x}_k$$

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